

Decision Trees

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Types of Classification Error

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training and Testing

Without opening the old lesson, explain one key idea from Training and Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Operational Labels

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Verify the candidate scores for Fitting a decision tree. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Fitting a decision tree. Proposed explanation: “A near branch must predict blocked”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Fit a new tree independently

New training rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked), (0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree. Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree predictions and compare accuracy with the training-majority baseline.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Build a tree for science books: first ask science?; then, for yes, pages<=100? Trace science/100 and history/70. What should happen to science with missing pages?

6. Solve the given case

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

Trace 90-page animal and science books.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A tree asks topic is space, then pages at most 60. A space book has 70 pages. Trace both answers and explain why you cannot stop at the first question.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 14 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Create a transparent tree for hard constraints before applying preference scores.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Build a tree for science books: first ask science?; then, for yes, pages $\leq$ 100? Trace science/100 and history/70. What should happen to science with missing pages?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Name the parts and decisions

Identify the root, its two branches, leaf predictions and depth of the worked fitted tree. Which parts were selected using training labels?

My reasoning and evidence

13. W2 Verify the candidate scores

For each near branch and each tall branch, count blocked and clear labels and choose a leaf. Give the total errors for each candidate question.

My reasoning and evidence

14. W3 Separate fitting from prediction

A learner sees a blocked validation case with $\text{near}=0$ and changes that leaf to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

My reasoning and evidence

15. W4 Fit a new tree independently

New training rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree. Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree predictions and compare accuracy with the training-majority baseline.

My reasoning and evidence

16. W5 Audit a tie and a claim

If near and tall make the same training error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Decision Trees

A fictional toy classifier predicts whether a book needs an illustration label.

Training rows (colour_cover,digram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or digram_inside; each leaf uses its training majority. A leaf tie chooses No. Select fewer training errors.

Decision Trees

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Colour candidate

Both colour leaves tie Yes/No, choose No: two training errors.



2. Diagram candidate

Diagram 1 leaf predicts Yes; diagram 0 leaf predicts No: zero errors.



3. Select tree

Choose the diagram question using the training comparison.



4. Apply new input

A new (1,0) row follows diagram=0 and predicts No.

CHANGE ONE THING

New input becomes (0,1).

Explain the changed result using the matching lesson.

Why does changing colour alone not change this tree's prediction?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and training-only selection evidence.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and training-only selection evidence.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy classifier predicts whether a book needs an illustration label.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A new (1,0) row follows diagram=0 and predicts No.

1. Select tree
2. Diagram candidate
3. Colour candidate
4. Apply new input

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Apply new input A new (1,0) row follows diagram=0 and predicts No. A question becomes a fitted tree only after its leaves and selection procedure are specified. The training comparison prefers the feature that separates these labels. Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

Practice 2: explain the deciding evidence

Changed-case practice: New input becomes (0,1). Which conclusion is supported by the supplied facts?

1. The fitted tree predicts Yes.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The fitted tree predicts Yes. It follows diagram=1; cover colour is not the selected question.

Repair a missing building block

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and prediction pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?. Return to this week and explain what you changed.

Review Training and Testing (ai-2/11). Goal: Choose a split unit that matches the intended prediction claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?. Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit evidence and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

